A network structure of emotional interactions in an electronic bulletin board

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Abstract

Recently, a large number of people use Social Networking Services (SNS). The study of SNS would reveal how atmosphere of our society changes as we interact through SNS. Here, we focus on emotions expressed in SNS and investigate how our emotions are affected by others in SNS. To reveal how positive/negative emotions are magnified and diffused through SNS, we analyze a network of emotional interactions based on a dictionary of emotional words for data of an electronic bulletin board.

Keyword: emotional interaction, social network, SNS

1. Introduction

Recently, we use various Social Networking Services (SNS), and the amount of data recording such human activities is increasing rapidly. The study of these data would reveal various aspects of our society. For example, the power law behavior is generally found in time evolution of the number of words appearing in the blogs as various social events take place [1].

One of the important characteristics of SNS is information transfer through various networks formed by people, materials and words. These social networks involve positive/negative aspects. As for negative aspects, groundless rumors and unlawful activities can spread very easily through SNS. Such cases would cause serious problems, since negative emotions tend to be more infectious than positive ones. Challenged by these problems, an experiment was done to influence negative/ positive expressions by artificially reducing the numbers of such expressions [2].

In our study, we focus on a network structure of emotional interactions in SNS. In particular, we analyze how positive/negative emotions are magnified and diffused through SNS based on the number of usages of positive/negative words.

2. Data description

In our study, we use data of an electronic bulletin board in Japanese online encyclopedia "Nico Nico Pedia" [3]. The data was provided by Mirai Kensaku Brazil Co., Ltd. through National Institute of Informatics in Japan.

For our analysis of Japanese sentences, we use the morphological analyzer "MeCab" [4]. It can divide sentences to words based on its own dictionary and assign part of speech to each word thus obtained.

For classification of emotional words, we use "Japanese Dictionary of Appraisal –attitude-" by Gengo Shigen Kyokai [5]. It enables us to classify emotional words according to whether they are positive or negative. It also provides their types based on the appraisal theory.

3. Data analysis

We analyze the relationship between messages and the responses to them. In the data, we choose those responses with a definite reference to the previous messages. Then, we count the number of emotional words contained in the original message and the responses to it. By counting the number of pairs of emotional words, we can construct a network between emotional words. We call it a network of emotional interactions. In particular, we study which emotional words form a dense network and how they provoke other's emotions. We will also investigate time intervals between messages and the responses to them containing pairs of emotional words.

References

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